

5(a)(i)	<i>structural gene: C ;</i> <i>control (regulatory) sequence: B ;</i> <i>repressor molecule: A ;</i>	3
5(a)(ii)	<i>any two from:</i> 1 is not, made all the time / constitutive ; 2 gene switched on / protein made, (only) when needed ; 3 triggered by a, change / stimulus / molecular signal / Msn2 / absence of glucose ; 4 concentration increases when galactose present ;	2
5(b)(i)	$\frac{226}{806} \times 100 ;$ 28 (%) ;	2

Question	Answer	Marks
5(b)(ii)	1 (Mig1 in nucleus) low(er) when glucose is absent ; ora 2 (as) Mig1 is not, needed in nucleus / bound to DNA ;	2
5(c)	any two from: lac Z / gene for β-galactosidase ; lac Y / gene for, lactose permease / β-galactoside permease ; lac A / gene for β-galactoside transacetylase ;	2